

Chapter 2 Exploring Data with Tables and Graphs Scatterplots, Correlation, and Regression (2.4)

Objectives

In this lesson, we will learn how to

- analyze paired sample data;
- use scatterplots to look for relationships between two quantitative variables;
- distinguish correlation from linear correlation;
- interpret the strength and direction of a linear relationship using the correlation coefficient r ;
- use a P-value to determine whether there is sufficient evidence of linear correlation;
- describe and use a regression line.

1. Paired Data

Many statistical questions involve two measurements obtained from the same individual or object.

Paired Data

Paired data consist of two related quantitative values for each observation.

They are usually written as

$$(x, y).$$

For example, for one person we might record

(shoe print length, height).

For one seal we might record

(overhead width, weight).

Our goal is often to determine whether the two variables are associated.

2. Correlation and Linear Correlation

Correlation

A correlation exists between two variables when the values of one variable are somehow associated with the values of the other variable.

Linear Correlation

A linear correlation exists when there is a correlation and the plotted points of paired data form a pattern that can be reasonably approximated by a straight line.

Correlation $\neq$ necessarily linear correlation

3. Scatterplots

Scatterplot

A scatterplot, or scatter diagram, is a graph of paired (x, y) quantitative data.

- The horizontal axis represents the first variable x .
- The vertical axis represents the second variable y .

- Each pair (x, y) is represented by one point.

Before calculating anything, we should usually

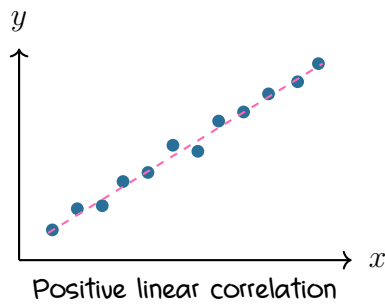
look at the scatterplot first.

Example 1: Weighing Seals with a Camera:

Suppose researchers record the overhead width and weight of several seals.

The scatterplot shows a clear upward pattern.

What does this suggest?



Solution

The distinct upward pattern suggests that the two variables are correlated.

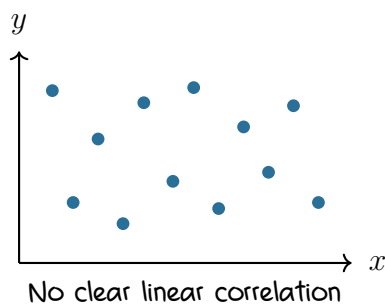
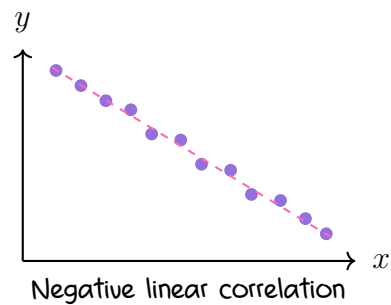
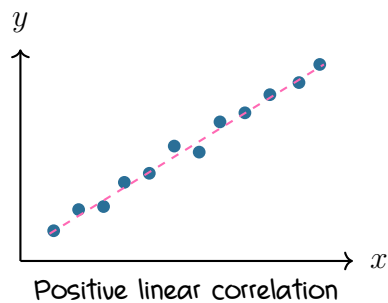
Seals with larger overhead widths tend to have larger weights.

This is an example of a

positive correlation.

4. Patterns in Scatterplots

A scatterplot can reveal several different types of patterns.



Direction of Linear Correlation

- **Positive correlation:** as x increases, y tends to increase.
- **Negative correlation:** as x increases, y tends to decrease.
- **No linear correlation:** the points do not show a clear straight-line pattern.

Example 2: Presidents and Opponents:

A scatterplot of presidents' heights and the heights of their main opponents shows no distinct pattern. What does this suggest?

Solution

The plotted points do not show a distinct pattern. Therefore, it appears that there is

no correlation

between the heights of presidents and the heights of their main opponents.

5. Linear Correlation Coefficient

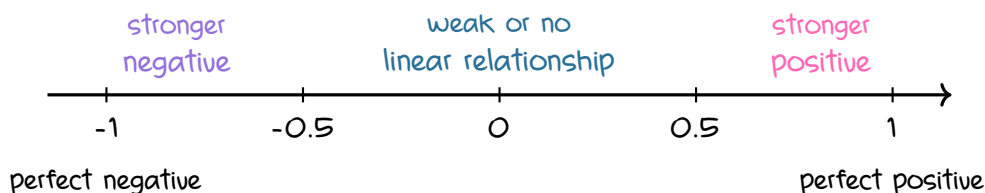
Looking at a scatterplot is useful, but we also want a numerical measure of the strength of a linear relationship.

Linear Correlation Coefficient

The linear correlation coefficient, denoted by r , measures the strength of the linear association between two variables.

The value of r is always between -1 and 1 :

$$-1 \leq r \leq 1.$$



Interpreting r

- If r is close to 1 , there appears to be a strong positive linear correlation.
- If r is close to -1 , there appears to be a strong negative linear correlation.
- If r is close to 0 , there does not appear to be a strong linear correlation.

Value of r	Interpretation
$r \approx 1$	Strong positive linear relationship
$r \approx -1$	Strong negative linear relationship
$r \approx 0$	Little or no linear relationship

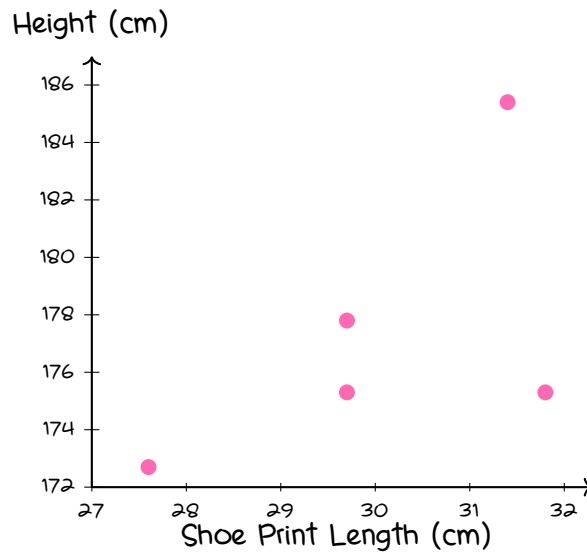
Important

The value r measures only a linear relationship.
A value near 0 does not necessarily mean that the variables have no relationship at all.

6. Example: Shoe Print Lengths and Heights

Consider the following five paired observations:

Shoe Print Length (cm)	29.7	29.7	31.4	31.8	27.6
Height (cm)	175.3	177.8	185.4	175.3	172.7



From the graph alone, it is not very clear whether there is a linear correlation.

Suppose technology gives

$$r = 0.591, \quad P\text{-value} = 0.294.$$

The value

$$r = 0.591$$

shows some positive linear association in the sample.

But is that enough evidence to conclude that a linear correlation exists in the population?

To answer that question, we consider the P-value.

7. P-Values for Linear Correlation

P-Value

If there really is no linear correlation between the two variables, the P-value is the probability of obtaining paired sample data with a correlation coefficient r at least as extreme as the one obtained from the sample.

For the five shoe-print observations,

$$P\text{-value} = 0.294.$$

This corresponds to

$$29.4\%.$$

That is a fairly high probability.

Example 3:

For the five shoe-print and height pairs,

$$r = 0.591, \quad P = 0.294.$$

Is there sufficient evidence to conclude that shoe print length and height have a linear correlation?

Solution

The P-value is

$$0.294.$$

This means that if there really is no linear correlation, there is a relatively high chance of obtaining a value of $r = 0.591$, or one even more extreme, from random sampling.

Because

$$0.294 > 0.05,$$

the P-value is not small.

Therefore,

There is not sufficient evidence to conclude that there is a linear correlation.

8. Interpreting a P-Value

Small P-Value

Only a small P-value, such as 0.05 or less, suggests that the observed sample result would be unlikely to occur by chance when there is no linear correlation.

Therefore, a small P-value supports a conclusion that there is a linear correlation.

A useful introductory guideline is

$$P \leq 0.05 \Rightarrow \text{sufficient evidence of linear correlation}$$

$$P > 0.05 \Rightarrow \text{not sufficient evidence of linear correlation}$$

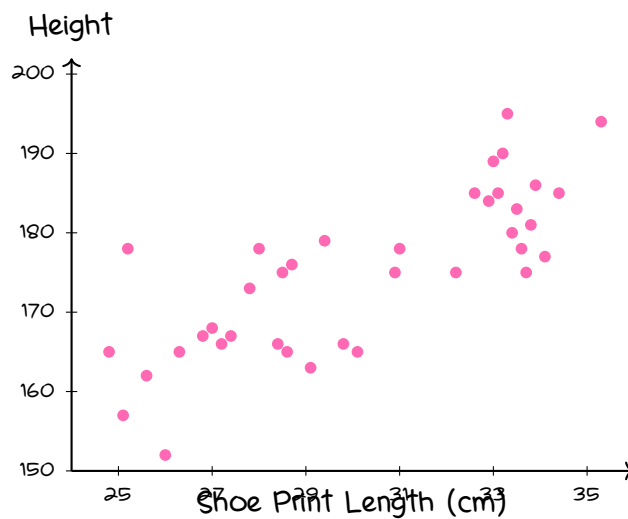
9. Larger Sample: Shoe Print Length and Height

Now suppose we have shoe-print lengths and heights for

$$n = 40$$

people.

The scatterplot has a much clearer pattern.



Technology gives

$$r = 0.813, \quad P\text{-value} = 0.000.$$

Example 4:
What should we conclude from these results?

Solution

The correlation coefficient is

$$r = 0.813.$$

This indicates a strong positive linear association in the sample.
Also,

$$P\text{-value} = 0.000,$$

which is very small.

Therefore, we have sufficient evidence to conclude that there is a linear correlation between shoe print lengths and heights.

Sample Size Matters

Compare the two examples:

Sample Size	r	P-value
5	0.591	0.294
40	0.813	0.000

A scatterplot, the value of r , and the P-value all contribute useful information when we assess linear correlation.

10. Regression

Once a useful linear relationship has been identified, we may want to use one variable to predict the other.

Regression Line

Given paired sample data, the regression line, also called the

- line of best fit, or
- least-squares line,

is the straight line that ``best'' fits the scatterplot.

The general regression equation is

$$\hat{y} = b_0 + b_1x,$$

where

b_0 = y -intercept,

b_1 = slope,

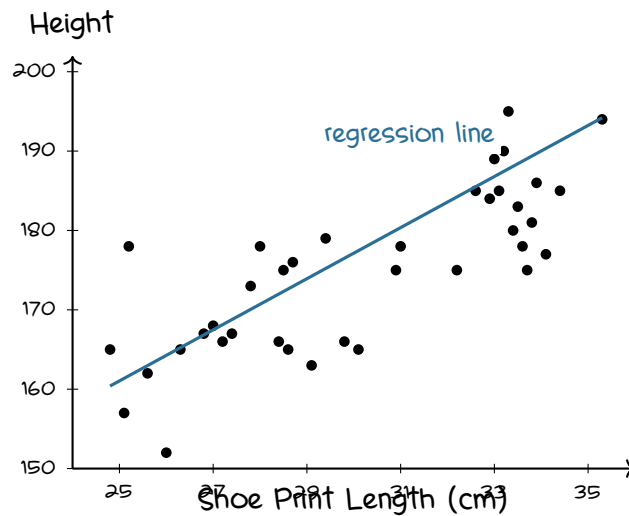
$\hat{y}$ = predicted value of y .

The symbol

$\hat{y}$

is read as

"y-hat."



For the larger shoe-print data set, the regression equation is

$$\hat{y} = 80.9 + 3.22x$$

or, using the variable names,

$$\text{Height} = 80.9 + 3.22(\text{Shoe Print Length}).$$

11. Using the Regression Equation

Example 5:

Use

$$\hat{y} = 80.9 + 3.22x$$

to predict the height of a person whose shoe print length is

30 cm.

Solution

Substitute

$$x = 30.$$

Then

$$\hat{y} = 80.9 + 3.22(30).$$

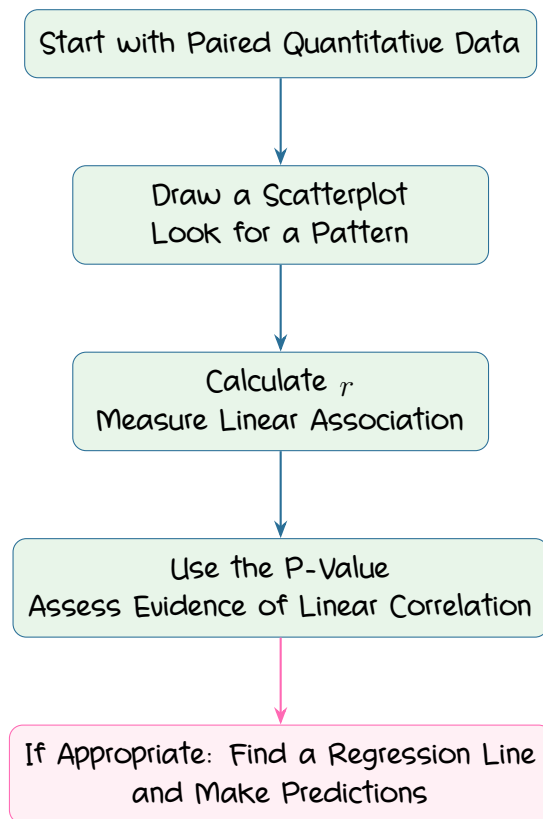
Therefore,

$$\hat{y} = 80.9 + 96.6 = 177.5.$$

So the predicted height is

$$\boxed{177.5 \text{ cm}}.$$

12. The Big Picture



13. Important Cautions

Correlation Does Not Prove Causation

A correlation between two variables does not by itself prove that changes in one variable cause changes in the other.

Use Regression Carefully

- Examine the scatterplot before relying on a regression line.
- Remember that r measures linear association.

- Predictions from a regression equation are estimates, not exact values.
- Be cautious about using a regression equation far outside the range of the observed data.

Summary

Key Ideas

- Paired data consist of two quantitative measurements associated with each observation.
- A scatterplot displays paired (x, y) quantitative data.
- Correlation means that two variables are associated.
- Linear correlation means the pattern can be reasonably approximated by a straight line.
- The linear correlation coefficient r measures the strength of linear association.
- The value of r always satisfies

$$-1 \leq r \leq 1.$$

- Values of r near 1 indicate positive linear association, while values near -1 indicate negative linear association.
- Values of r near 0 indicate little or no linear association.
- A small P-value supports a conclusion that there is a linear correlation.
- A regression line is the straight line that best fits the paired sample data.

- The regression equation has the form

$$\hat{y} = b_0 + b_1x.$$